



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD MPP.7

Math is Playful

Lesson 10-3 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

Language Objective: I can describe a puzzle strategy using the words pattern, rule, and predict.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

puzzle

pattern

pattern rule

predict

organize



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— A problem designed to be played with — it has rules, a goal, and room to imagine solutions.

2

— Something that repeats or changes in a predictable way.

3

— A rule that tells you how to get the next value in a pattern from the one before it.

4

— To say what will happen before it happens, using a pattern or reasoning.

5

— To arrange information — often in a table — so patterns become easier to see.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How can a pattern help you imagine the solution?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **puzzle** — A problem designed to be played with — it has rules, a goal, and room to imagine solutions.
rompecabezas — Un problema diseñado para jugarse: tiene reglas, una meta y espacio para imaginar soluciones.

☐ **pattern** — Something that repeats or changes in a predictable way.
patrón — Algo que se repite o cambia de una manera predecible.

☐ **pattern rule** — A rule that tells you how to get the next value in a pattern from the one before it.
regla del patrón — Una regla que te dice cómo obtener el siguiente valor de un patrón a partir del anterior.

☐ **predict** — To say what will happen before it happens, using a pattern or reasoning.
predecir — Decir lo que va a pasar antes de que pase, usando un patrón o el razonamiento.

☐ **organize** — To arrange information — often in a table — so patterns become easier to see.
organizar — Ordenar la información, muchas veces en una tabla, para que los patrones sean más fáciles de ver.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The pattern rule for the Tower of Hanoi is to ____ the number of steps for the smaller tower and ____ for each new disc. Applying this rule twice starting from 7 steps for a three-disc tower gives a five-disc tower a minimum of ____ steps.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The pattern rule is to double the previous number of moves and add 1: $1 \times 2 + 1 = 3$, and $3 \times 2 + 1 = 7$.

Evidence: A strong answer states 'double and add 1' and checks it against both jumps (1→3 and 3→7). A weak answer only notices the numbers are growing without naming the rule.

Reasoning: This evidence connects the definition of puzzle to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The pattern rule for the Tower of Hanoi is to ____ the number of steps for the smaller tower and ____ for each new disc. Applying this rule twice starting from 7 steps for a three-disc tower gives a five-disc tower a minimum of ____ steps.

- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- I know because ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** puzzle
2. **Notes 2:** pattern
3. **Notes 3:** pattern rule
4. **Notes 4:** predict
5. **Notes 5:** organize
6. **Watch for this mistake:** *Doubling the previous number of steps but forgetting to add 1 — answering 14 instead of 15 for four discs, or 30 instead of 31 for five.*
7. **Try It 1:**
8. **Try It 2:**